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1
2 import java.io.FileNotFoundException;
3 import java.io.File;
4 import java.util.*;
5 import java.text.DecimalFormat;
6
7 public class AIChatBotGenerator_State
8 {
9
10     static Scanner sc = new Scanner(System.in); //SC2
11     static ArrayList<ChatBots> chat_Bots = new ArrayList<ChatBots>(); //SC2
12     public static void main (String args [])
13     {
14
15         inputManager(); //Given
16
17         printChatBots();
18         printHighestComplexity(); //Given
19         printLowestComplexity(); //Given
20         searchByLanguage();
21     }
22
23
24
25
26 //ChatBots will create. Signature only
27 private static void printHighestComplexity() //ArrayList<ChatBots> stuList
28 {
29     int index = 0;
30     double number = 0;
31
32     for(int i = 1; i < chat_Bots.size(); i++)
33     {
34         if(chat_Bots.get(i).getComplexity()>chat_Bots.get(index).getComplexity()) //SC3
35         {
36             index = i;
37             number = chat_Bots.get(i).getComplexity();
38         }
39     }
40
41     System.out.println("\n\nMost Expensive Complexity Rate: "+ chat_Bots.get(index).getComplexity()+ " "+chat_B
ots.get(index).getSupportedLanguages());
42 }
43
44 //STATE POINT
45 //ChatBots will create. Signature only
46 private static void printLowestComplexity()
47 {
48     int index = 0;
49     double number = 0;
50
51     for(int i = 1; i < chat_Bots.size(); i++)
52     {
53         if(chat_Bots.get(i).getComplexity()<chat_Bots.get(index).getComplexity()) //SC4
54         {
55             index = i;
56             number = chat_Bots.get(i).getComplexity();
57         }
58     }
59
60     System.out.println("\n\nLeast Expensive Complexity Rate: "+ chat_Bots.get(index).getComplexity()+ " "+chat_
Bots.get(index).getSupportedLanguages());
61 }
62
63 //Given
64 private static void setChatBots(int bC)

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67 {
68     Random random = new Random();
69     ChatBots cB;
70     DecimalFormat df = new DecimalFormat("#.0000");
71     int botCount = bC;
72     double complexity;
73     int token_Rate;
74     String native_Language; String secondary_Language;
75
76     String [] supportedLanguages ={"English", "French", "Chinese", "Spanish", "Arabic", "Russian", "German", "J
apanese", "Portuguese", "Hindi", "Italian", "Dutch", "Korean", "Turkish", "Vietnamese", "Swedish", "Polish", "Indones
ian", "Greek", "Hebrew", "Thai", "Czech", "Romanian", "Hungarian", "Finnish"};
77
78     for(int i =0; i<botCount; i++)
79     {
80
81         do{
82
83             native_Language = supportedLanguages[random.nextInt(supportedLanguages.length)];
84             secondary_Language = supportedLanguages[random.nextInt(supportedLanguages.length)];
85             }while(native_Language.equals(secondary_Language));
86             token_Rate = random.nextInt(16)+5;
87             double tempComplexity = random.nextDouble()*7.1 + 2.0;
88             if(tempComplexity > 9.0) tempComplexity = Math.floor(tempComplexity);
89             String tempString =df.format(tempComplexity);
90             complexity = Double.parseDouble(tempString);
91             cB = new ChatBots(native_Language,secondary_Language,token_Rate,complexity);
92             chat_Bots.add(cB);
93         }
94     }
95
96     //Given
97     /*
98     private static void printChatBots(){
99         int i = 1;
100         for (ChatBots c : chat_Bots)
101         {
102             System.out.println(i+" "+c.getSupportedLanguages() + " | Token Rate: "+c.getTokenRate()+" | Complexity:
"+c.getComplexity());
103             i++;
104         }
105     }
106     printHighestComplexity();
107     printLowestComplexity();
108 }
109 */
110 private static void printChatBots() { //SC11
111     System.out.println("-----");
112     System.out.printf("| %-4s | %-30s | %-10s | %-10s |\n", "No.", "Supported Languages", "Token Rate", "Complexi
ty"); //SC11
113     System.out.println("-----");
114     int i = 1;
115     for (ChatBots c : chat_Bots) {
116         System.out.printf("| %-4d | %-30s | %-10d | %-10.4f |\n", i, c.getSupportedLanguages(), c.getTokenRate(),
c.getComplexity()); //SC11
117         i++;
118     }
119     System.out.println("-----");
120     printHighestComplexity();
121     printLowestComplexity();
122 }
123
124 private static void inputManager()
125 {
126     int temp =0;

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```

130     int token_Rate =0;
131     double complexity =0.0;
132     String native_Language;
133     String secondary_Language;
134     String tempString;
135     ChatBots cB;
136     DecimalFormat df = new DecimalFormat("#.0000"); //SC9
137
138
139     while(true){
140         try{ //SC5
141             do{
142                 temp =0;
143                 System.out.println("_____");
144                 System.out.println("Welcome to the BPA Chatbot Generator. Choose one of the options:");
145                 System.out.println("Press [1] to create random chatbots OR Press [2] to create a single chatbot \n
OR Press [3] to search by language OR Press [4] to end the program");
146                 temp = sc.nextInt();
147                 if(temp <= 0 || temp > 4) System.out.println("Your entry is out of range."); //SC5
148                 sc.nextLine();
149             } while(temp <= 0 || temp > 4);
150             if(temp ==1){
151                 do{
152                     temp =0;
153                     try{ //SC6
154                         do{
155                             System.out.print("Please enter in a value between 1 and 50: ");
156                             temp = sc.nextInt();
157                             if(temp >50 || temp <1) System.out.println("Your entry is out of range."); //SC6
158                             sc.nextLine();
159                         }while(temp>50|| temp<1);
160                     }
161                     catch(InputMismatchException e) //SC6
162                     {
163                         sc.next();
164                         System.out.println("\nERROR: Please enter valid data type.");
165                     }
166                 } while(temp >50 || temp <1);
167                 setChatBots(temp);
168                 temp =0;
169                 System.out.println("_____");
170                 printChatBots();
171             }
172             if(temp ==2){
173
174                 System.out.println("You are now creating a single chatbot");
175                 System.out.println("Enter in the native language");
176                 native_Language = sc.nextLine().toLowerCase();
177
178                 System.out.println("Enter in the secondary language");
179                 secondary_Language = sc.nextLine().toLowerCase();
180                 do{
181                     try{ //SC7
182                         do{
183                             System.out.println("Enter the token rate value (5 to 20): ");
184                             token_Rate = sc.nextInt();
185                             if(token_Rate >20 || token_Rate <5) System.out.println("Your entry is out of range."); //
SC7
186                         }while(token_Rate >20 || token_Rate <5);
187                     }
188                     catch(InputMismatchException e) //SC7
189                     {
190                         sc.next();
191                         System.out.println("\nERROR: Please enter valid data type.");
192                     }
193                 }while(token_Rate >20 || token_Rate <5);
194                 do{
195                     try{ //SC8

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196         do{
197             System.out.println("Enter in the token complexity in its proper format (#####) in a range
of 2.0 to 9.0");
198             complexity = sc.nextDouble();
199             tempString = df.format(complexity); //SC9
200             complexity = Double.parseDouble(tempString);
201             if(complexity>9.0 || complexity < 2.0) System.out.println("Your entry is out of range.");
//SC8
202             }while(complexity>9.0 || complexity < 2.0);
203         }
204         catch(InputMismatchException e) //SC8
205         {
206             sc.next();
207             System.out.println("\nERROR: Please enter valid data type.");
208         }
209         }while(complexity>4.0 || complexity <= 0.0);
210         native_Language = native_Language.substring(0,1).toUpperCase() + native_Language.substring(1).toLo
werCase(); //SC10
211         secondary_Language = secondary_Language.substring(0,1).toUpperCase() + secondary_Language.substrin
g(1).toLowerCase(); //SC10
212         cB = new ChatBots(native_Language,secondary_Language,token_Rate,complexity);
213         chat_Bots.add(cB);
214         System.out.println("_____");
215         printChatBots();
216     }
217     if(temp ==3 ){
218         if (chat_Bots.size()==0){
219             System.out.println("You must create chatbots first.");
220         } else {searchByLanguage();}
221     }
222 }
223
224     if(temp == 4){
225         System.out.println("Goodbye!");
226         System.exit(0);
227     }
228 }
229 catch(InputMismatchException e) //SC5
230 {
231     sc.next();
232     System.out.println("\nERROR: Please enter valid data type.");
233 }
234 }
235 }
236 }
237 private static void searchByLanguage() {
238     System.out.println("Enter the language you want to search for:");
239     String searchLanguage = sc.nextLine().toLowerCase();
240
241     boolean found = false;
242     for (ChatBots bot : chat_Bots) {
243         if (bot.getSupportedLanguages().toLowerCase().contains(searchLanguage)) {
244             System.out.println("Chatbot found: " + bot.getSupportedLanguages() + " | Token Rate: " + bot.getToken
Rate() + " | Complexity: " + bot.getComplexity());
245             found = true;
246         }
247     }
248
249     if (!found) {
250         System.out.println("No chatbot found for the specified language.");
251     }
252 }
253
254 }
255
256 //SC1
257 class ChatBots
258 {

```

```

259 //Create appropriately variables based upon constructor parameters
260 String native_Language;
261 String secondary_Language;
262 int token_Rate;
263 double complexity;
264 //Generic Constructor
265 public ChatBots()
266 {
267     native_Language = "Esparanto";
268     secondary_Language = "Java";
269     token_Rate = 0;
270     complexity = 0.0;
271 }
272 //Constructor with Parameters
273 public ChatBots(String nL, String sL, int tR, double co)
274 {
275     native_Language = nL;
276     secondary_Language = sL;
277     token_Rate = tR;
278     complexity = co;
279 }
280
281 //Returns "last name, first name" with appropriate variables
282 public String getSupportedLanguages()
283 {
284     return secondary_Language + ", "+native_Language;
285 }
286 //Returns first name
287 public String getNativeLanguage()
288 {
289     return native_Language;
290 }
291 //Returns last name
292 public String getSecondaryLanguage()
293 {
294     return secondary_Language;
295 }
296 //Returns the Complexity
297 public double getComplexity()
298 {
299     return complexity;
300 }
301 //Returns token_Rate level
302 public int getTokenRate()
303 {
304     return token_Rate;
305 }
306
307
308 }

```